Intelligent Property Management Workflow with OCR-Enabled Automated Billing Classification and Tenant Notification

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CHAPTER I

THE PROBLEM AND ITS BACKGROUND

This chapter presents the background and related literature of the study entitled “Intelligent Property Management Workflow with OCR-Enabled Automated Billing Classification and Tenant Notification.” It also presents the significance of the study, the statement of the problem, and the scope and limitations of the system.

1.1 Introduction

Property management involves maintaining tenant records, managing utility bills, computing charges, monitoring payment status, and providing billing information to tenants. These activities require the organized handling of property records, tenant information, billing documents, and other administrative records. Cammarata (2023) discusses the importance of records and information management in real-estate operations, while Teo et al. (2024) examine property-management functions and services. When property managers handle these activities through separate or manual methods, organizing information and maintaining consistent work processes can become more difficult.

To support these activities, digital approaches provide a way to centralize property information and organize regular management tasks. Property Technology (PropTech) refers to the use of digital technologies in real-estate and property-management



activities. Siniak et al. (2020) discuss the role of PropTech in changing real-estate operations and services, while Oyetunji et al. (2025) examine digital solutions for streamlining property-management operations in affordable-housing communities. Their discussion highlights how digital tools may support routine management activities, resource allocation, and tenant interaction, although the study also notes barriers such as financial constraints, infrastructure gaps, and resistance to change. Overall, this literature shows how digital technologies support property and housing-management activities and provides a basis for using centralized systems to organize property information and management processes.

One important area in property management is billing. Utility billing requires information from billing documents before charges can be processed, including amounts, dates, billing periods, providers, and account or reference identifiers. Optical Character Recognition (OCR) provides a way to extract text and relevant information from supported documents such as receipts and invoices (Kumar et al., 2020). Saout et al. (2024) discuss invoice-data extraction as a process involving document digitization, OCR, information extraction, and recognition of document structures. Mohan (2023) also presents a staged pipeline for extracting key-value information from commercial invoices. Overall, these studies support the use of OCR and document-processing methods to extract and organize information from billing documents for later processing.

After billing information is extracted, it still needs to be validated, classified, and processed according to defined billing conditions. Rule-based processing uses

predefined conditions and logical rules to support structured operational decisions and information handling. Baralis et al. (2022) discuss business-rule engines that represent and execute operational decisions in a structured way. In relation to process automation, Geyer-Klingeberg et al. (2024) present a rule-based method that analyzes decision logic and classifies business processes according to their suitability for robotic process automation. These ideas are relevant to a billing workflow in which validation, classification, calculation, and exception-handling conditions are defined in advance. The reviewed literature provides a basis for combining OCR, rule-based processing, and notification technologies in a billing workflow for smaller residential property-management settings.

Based on these ideas, the study proposes a web-based workflow that helps property managers organize billing information and allows tenants to view their bills and history. Its main concept is to receive utility bills, extract information through OCR, and use predefined rules to process valid records or identify those requiring Manager review. In this study, “intelligent” refers to OCR-assisted extraction and rule-based processing rather than autonomous decision-making. The workflow separates Manager-confirmed rent payments from personal utility tracking and provides permitted Web Push alerts that direct tenants to the authenticated portal.

1.2 Review of Related Literature

Digital Property Management and PropTech

Digital property management increasingly relies on technology to organize property details, management activities, billing data, and related services. Several studies have examined this shift from different perspectives: Teo et al. (2024) discuss technology and digitalization in property-management functions and service delivery, while Ding and Sun (2024) review technology innovations in housing management. In this context, PropTech refers to the use of digital technologies in the real estate and property-management sector. Siniak et al. (2020) examine the impact of PropTech on real estate industry growth, Saputra et al. (2023) review digital transformation trends in the property-management industry, and Al-haimi et al. (2025) synthesize the technologies, benefits, and challenges associated with digital transformation in real estate. In addition, Cammarata (2023) discusses the importance of records and information management in managing risks during real-estate changes. Overall, this literature supports the relevance of centralized digital tools for property information and administrative processes. Building on this literature, the present study proposes a centralized web-based workflow that organizes property, unit, tenant, utility, billing, payment, exception, and receipt information for selected boarding houses and residential apartments.

OCR and Billing Document Extraction

Within property-management operations, billing is one area where digital processing can support routine administrative work. Billing documents contain information such as amounts, dates, billing periods, providers, and account or reference identifiers. Kumar et al. (2020) examine the extraction of information from bill receipts using Optical Character Recognition (OCR), while Saout et al. (2024) provide an overview of invoice-data extraction involving document digitization, OCR, information extraction, and document-structure recognition. Similarly, Mohan (2023) presents an end-to-end pipeline for custom key-value extraction from commercial invoices. However, extracting information from a billing document is only the first stage of billing processing. In the proposed system, OCR is used to process supported standard digital PDFs and clear images. After extraction, the information undergoes validation before billing processing. Records that do not meet the required validation conditions are sent to the Exception Queue for Manager review. Thus, OCR is treated as a method for extracting information from billing documents rather than as a guarantee that every extracted value is accurate. The OCR component may employ machine-learning-based text-recognition technology to convert visible characters and document text into machine-readable output. However, the proposed system does not use OCR output or machine-learning recognition to determine billing validity, classify payment status, compute charges, or make property-management decisions.

Rule-Based Billing Processing

Once billing information has been extracted, it must be checked and processed according to defined conditions. Rule-based processing uses explicit conditions and decision rules to classify, validate, and route information. Wu et al. (2022) demonstrate the use of rule-based information extraction to organize domain-specific information, while Baralis et al. (2022) discuss business-rule engines for representing and executing operational decisions in a structured manner. Extending this idea to process automation, Geyer-Klingeberg et al. (2024) present a rule-based method that analyzes decision logic and classifies business processes according to their suitability for robotic process automation, while Kim et al. (2011) examine proactive exception handling in business processes. These studies support the use of predefined and traceable conditions to determine whether a record proceeds through normal processing or requires additional review. Applying these principles, the proposed system uses defined rules for billing validation, provider and account/reference mapping, duplicate detection, and identifying individual or shared billing arrangements. Records that do not meet the defined conditions are sent to the Exception Queue for Manager review. These rules, together with the separation between Tenant-reported rent and Manager-confirmed rent status, are system requirements rather than findings established by the cited studies. Digital rent receipts are created only after Manager confirmation. Personal utility tracking remains separate and does not establish provider settlement.

Shared Utility Billing and Cost Allocation

Shared utility billing presents a related issue because a single charge may need to be allocated among several occupants. Levinson and Niemann (2004) examine apartment energy use when landlords pay for utilities, while Mayr and Souza (2025) discuss coordination and incentive issues associated with shared heating costs. These studies provide context for understanding cost responsibility and consumption incentives in shared-utility arrangements. Nevertheless, they do not establish one universal allocation formula for rental properties or determine how the proposed system should calculate individual tenant charges. For this reason, the proposed system treats the allocation method as a defined system requirement rather than as a formula established by the reviewed literature. For shared occupancy, charges are split equally among eligible tenants in a Manager-configured group. The billing requirements also include reconciliation controls so that allocated charges equal the source billing amount. The allocation formula, rounding procedure, and reconciliation method are described in the succeeding chapters and will be tested during system evaluation.

Web Push Notifications and Secure Notification Delivery

After a billing record is finalized, tenants need to be informed that their billing information is available. The World Wide Web Consortium (2024a) describes the Push API, including push services and user subscriptions. The World Wide Web Consortium (2024b) also explains the role of service workers in handling web events in the background when they are included in the system implementation. Because Web Push involves browser subscriptions and authorization, security must also be considered.

Carboneri et al. (2023) examine Web Push implementations in the wild and identify security concerns involving push subscriptions and authorization. In the proposed system, Web Push Notifications serve as billing alerts that direct tenants to the authenticated web portal, where complete billing information is available. However, Web Push does not confirm payment, prove that payment occurred, or show whether the tenant read the notification. Tenant payment reporting therefore remains separate from Manager confirmation.

Iterative Software Development

Because the proposed system contains several connected functions, its development requires an organized and adaptable process. Iterative software development divides development into repeated cycles of building, reviewing, receiving feedback, and improving. Larman and Basili (2003) explain the development of iterative and incremental approaches, in which software is developed progressively through repeated cycles. Inayat et al. (2015) emphasize stakeholder involvement, communication, and requirements refinement in iterative requirements engineering, while Ramesh et al. (2021) identify challenges involving changing requirements, stakeholder participation, documentation, and communication in Agile software development. These studies support the use of Iterative Software Development Methodology in the present study. Although the literature supports this approach, the actual quality and performance of the proposed system will be determined through implementation, testing, user feedback, and evaluation.



Software Quality Evaluation and Security

To evaluate the proposed system systematically, the study requires defined software-quality criteria. ISO/IEC 25010:2023 provides a product-quality model for systems and software quality requirements and evaluation (International Organization for Standardization, 2023). Guided by this model, the study will evaluate five quality characteristics: Functional Suitability, Performance Efficiency, Interaction Capability, Reliability, and Security. These characteristics provide a basis for examining whether the system performs its intended functions, performs efficiently, supports user interaction, operates consistently, and protects information. Security is particularly important because the system handles tenant and billing information. Republic Act No. 10173, or the Data Privacy Act of 2012, requires organizations that process personal information to use reasonable and appropriate measures to protect it (Republic of the Philippines, 2012). The National Privacy Commission (2016) further discusses safeguards for the confidentiality, integrity, and availability of personal data, while Zafar et al. (2022) review authorization and access-control strategies that restrict system functions and information according to user permissions. In the proposed system, role-based access separates Manager and Tenant functions. These sources guide the study's quality and security evaluation, but they do not prove that the system already meets the selected quality characteristics or privacy requirements. The Security evaluation is limited to the defined system functions, access controls, and security-related criteria included in the study's evaluation procedures. It does not constitute a legal audit, certification, or claim of complete compliance with the Philippine Data Privacy Act.

1.3 Significance of the Study

The study develops and evaluates a web-based property-management workflow that uses OCR-based information extraction, rule-based billing classification, and Web Push Notifications. It primarily benefits property managers and tenants and may serve as a reference for future researchers and system developers.

In the global context, the study contributes an applied example of PropTech and digital property management by showing how existing technologies can be combined into one workflow for residential property operations. It also provides a local example of how digital property-management concepts can be applied in a smaller property-management setting. The study does not claim that the proposed workflow represents all property-management environments; rather, its significance lies in demonstrating how selected technologies may be integrated within the defined scope of the study.

In the economic context, the system is designed to reduce repetitive manual activities involved in receiving utility bills, recording billing information, calculating tenant charges, and communicating billing updates. Centralized records and predefined billing rules can also help property managers follow a more consistent billing process. Processing time will be assessed during system evaluation. The study does not establish reductions in operating costs or late payments.

In the environmental context, the system uses electronic utility bills, digital billing records, and electronic notifications. This can reduce the need for paper-based records during the billing process. The study does not claim a specific reduction in paper use because this is not directly measured.

In the societal context, the system provides property managers and tenants with organized billing information. Itemized billing records, defined billing rules, and automated notifications can help make tenant dues easier to understand and communicate. This can support clearer billing transactions between property managers and tenants. However, the study does not claim that the system will eliminate billing misunderstandings or disputes because those outcomes are not directly measured.

1.4 Statement of the Problem

This study develops and evaluates an Intelligent Property Management Workflow with OCR-Enabled Automated Billing Classification and Tenant Notification for selected boarding houses and residential apartments in Pasig City.

Specifically, the study seeks to answer the following questions:

1. What development stages are undertaken in developing the proposed system using the Iterative Software Development Methodology?

2. How does the proposed system integrate OCR, rule-based billing processing, exception handling, payment management, digital receipts, and Web Push Notifications into one property-management workflow?

3. How does the developed system perform when evaluated using the selected ISO/IEC 25010:2023 quality characteristics in terms of:

    a. Functional Suitability;

    b. Performance Efficiency;

    c. Interaction Capability;

    d. Reliability; and

    e. Security?

4. What implementation guide and user manual can be prepared for the developed system?

1.5 Scope and Limitations of the Study

The study will develop and evaluate a web-based property-management workflow for selected boarding houses and residential apartments in Pasig City during Academic Year 2026-2027. It will centralize property, unit, tenant, utility-mapping, rent, billing, payment, exception, correction, receipt, and notification records. “Intelligent” refers to

OCR text extraction combined with explicit, deterministic, and traceable processing rules. An existing OCR engine may use machine-learning-based recognition, but the researchers will not train or customize a model. Validation, classification, computation, allocation and payment decisions will follow predefined rules and Manager-controlled actions rather than autonomous AI decisions.

Billing will cover rent, electricity, and water only. Rent will be fixed and non-prorated, generated once per monthly anniversary cycle with the same due date. Short months will use their final day while preserving the original anniversary day for later months. After an outage, missed eligible cycles will be generated once with their original due dates and actual creation dates retained. The first bill will show current rent and one-month advance separately. Confirmed advance will be reserved for the final applicable rent cycle and applied only with Manager confirmation, using the original payment and receipt without another cash transaction or duplicate receipt. Applied credit cannot be reused, and no replacement advance will be charged automatically. Individual charges will follow the configured lease or utility account. Shared charges will be split equally among eligible tenants in the Manager-configured group, with two-decimal shares totaling the source charge. Tenants will choose the payer of a shared provider bill outside the system; allocation will not represent contribution collection or verified provider settlement.

The system will support month-to-month rentals that continue until the Tenant leaves. The Manager will record the rental start date and leave the end date blank while

occupancy continues. The end date will be recorded when the Tenant leaves. Rent will follow the monthly anniversary rule, with no new cycles generated after the recorded end date. The existing Tenant account will retain access to its own bills, utility tracking history, confirmed payments, and receipts, and unpaid balances will not be removed merely because occupancy has ended. Departed tenants may report outstanding rent payments for Manager confirmation. Advance rent will equal one month of the Tenant's configured rent or, for shared rent, the Tenant's calculated rent share at the time the advance is charged. Its amount will not be entered as an arbitrary credit allowance. A later change in rent or shared allocation that causes a credit mismatch will require Manager review. Automatic unit transfers and lease-renewal processing will not be included in the study.

Utility bills will enter the system through attachments received by the designated property-management email account or through optional Manager uploads of clear photographs or scans of printed provider bills. The Manager may hand a printed bill directly to tenants without uploading it. Until an email or uploaded copy is received and validated, that bill will not be available in the portal for automatic allocation, personal tracking, history, or reminders. OCR will extract the amount due, due date, billing period, provider, and account or reference identifier from supported digitally generated PDFs and clear images of selected provider layouts. Required-field, mapping, period, and duplicate checks will precede classification and computation, including checks across email and uploaded copies of the same bill. Incomplete, uncertain, duplicated, unmappable, or otherwise invalid records will enter the Exception Queue for Manager correction, revalidation, and reprocessing. Finalized corrections will preserve prior values

and tracking history. Utility due dates will come from validated received documents; the system will not predict the next billing period's due date.

Rent payment statuses will remain Unpaid, Tenant Reported Paid, and Paid. Tenants may report rent payments for Manager review. The Manager may also record and confirm an actual rent payment without a prior Tenant report, after checking it outside the system. Only the Manager may authorize confirmed rent receipts; rejected reports will return to Unpaid with a recorded reason. Utilities will use a separate Tenant-controlled tracker with Not Marked Paid and Self-Reported Paid, accompanied by “Provider payment not verified.” A mark will refer only to the Tenant's own utility bill or allocated share and will neither require Manager confirmation nor generate a system-issued utility receipt. For shared bills, one Tenant's mark will not mark other shares or the whole provider account as paid. The system will not independently verify payments. Tenants will access their billing history, rent statuses, utility tracking records, and confirmed rent receipts through the authenticated portal. Web Push reminders will be scheduled at 9 AM Philippine time, three calendar days before the actual due date, for eligible Unpaid rent and utility records Not Marked Paid. Pending rent reports and utilities marked Self-Reported Paid will be excluded. Bills unavailable before the scheduled reminder time, including rent generated on its due date, will receive billing updates rather than backdated reminders. Push permission, device support, or delivery failure will not prevent portal access, and an alert will not prove payment or readership. Tenant SMS and email notifications are excluded.

Admin access will be limited to system maintenance. The Manager will manage property and tenant records, utility mappings, uploaded bills, billing rules, exceptions, corrections, rent payment confirmation, advance-credit application, and confirmed rent receipts. Tenants will view their permitted records, report rent payments, maintain their own utility tracking marks, and receive permitted alerts. Authentication and role restrictions will protect centralized records. Evaluation will cover Functional Suitability, Performance Efficiency, Interaction Capability, Reliability, and Security under the selected ISO/IEC 25010:2023 characteristics. It will not establish long-term changes in revenue, profitability, payment compliance, or Tenant behavior.

The study excludes mobile applications; installation or physical reading of smart meters or submeters; Internet or Wi-Fi billing; predictive profit analytics or billing; Generative AI and autonomous AI billing decisions; blockchain; automatic utility disconnection; payment gateways or bank integration; independent external payment verification; maintenance tickets; legal contract generation; and automated eviction. OCR will not cover handwritten, heavily damaged, severely distorted, or low-resolution documents. The study does not guarantee notification delivery or readership, establish a universally fair utility-allocation formula, or provide complete legal certification under the Philippine Data Privacy Act.

CHAPTER II

METHODOLOGY

This chapter presents the methodology and procedures that will guide the development of the proposed Intelligent Property Management Workflow with OCR-Enabled Automated Billing Classification and Tenant Notification. It describes the project requirements, system design, conceptual operation, interface mockups, and Iterative Software Development Methodology. It also presents the testing and evaluation procedures that will be used to examine the developed system within the approved scope of the study.

2.0 Project Overview

Project Information

The project, entitled Intelligent Property Management Workflow with OCR-Enabled Automated Billing Classification and Tenant Notification, is a web-based system for selected boarding houses and residential apartments in Pasig City. It supports property, unit, tenant, rent, electricity, water, utility-mapping, billing, payment-status, exception, receipt, and applicable notification records. The system provides an integrated workflow for rent and utility billing. It receives utility-bill attachments through the designated property-management email account and allows optional Manager uploads of clear photographs or scans of printed provider bills. OCR extracts available

information, which is validated and processed through predefined rules before billing computation and finalization. The system manages exceptions, records rent payment reports and Manager confirmations, supports approved use of confirmed advance credit, and provides digital receipts for confirmed rent payments. Electricity and water use separate personal tracking marks maintained by the assigned Tenant, without provider payment verification or system-issued utility receipts. Permitted Web Push billing updates and due-date reminders direct tenants to the authenticated portal.

In this study, intelligent refers to OCR-enabled document text extraction combined with predefined, explicit, deterministic, and traceable rules. The OCR component may use machine-learning-based text recognition to extract available document information, but it does not make autonomous billing, validation, allocation, payment, or property-management decisions. These functions use defined rules and Manager-controlled actions.

Requirements Definition

The proposed system is designed to meet the following business, user, and security requirements.

A. Business Requirements

The system shall retrieve supported utility-bill attachments from the designated property-management mailbox and allow optional Manager uploads of clear photographs or scans of printed provider bills. OCR shall extract the amount due, due date, billing



period, provider, and account or reference identifier. Explicit, deterministic, and traceable rules shall check required values, provider/account mappings, periods, and duplicates across both intake methods, identify the billing category and individual or shared arrangement, and route unprocessable records to the Exception Queue. Only valid records shall proceed to computation and finalization. Manager corrections shall undergo revalidation and, where applicable, reprocessing or recomputation, including corrections to finalized records.

Billing shall cover rent, electricity, and water. Fixed, non-prorated rent shall be generated once per lease cycle on the anniversary date, using the last calendar day when necessary and retaining the original day for later months. Missed eligible cycles shall be recovered once after server availability resumes, with original due dates and actual creation dates retained. One-month advance rent shall appear separately on the first bill and become available credit after payment confirmation. Only the Manager shall confirm applying available credit to the final applicable rent cycle. Applied amounts shall not be reused, and the system shall not automatically create a replacement advance charge. Shared charges shall be split equally among eligible tenants in the configured group, using the scheduled rent-cycle date for rent and the final day of the source period for utilities. For recovered rent cycles, eligibility shall use the original scheduled date rather than the recovery date. The eligible list shall be stored. Whole-centavo computation and remainder distribution in ascending permanent Tenant-record identifier order shall preserve the source total. Missing groups or eligible tenants shall require exception review. Duplicate billing, repeated advance-rent lines, and repeated credit

application shall be prevented. Tenants shall arrange who pays a shared provider bill outside the system; calculating shares shall not establish provider settlement.

The centralized database shall retain property, unit, Tenant, mapping, rent, billing, rent-payment, utility-tracking, exception, correction, receipt, and applicable notification records. Rent bills shall use Unpaid, Tenant Reported Paid, and Paid. Tenant reports shall remain unconfirmed until Manager confirmation; the Manager may also record and confirm an actual rent payment without a prior report after external checking. Receipts shall be issued only for confirmed rent payments. Utilities shall use separate Not Marked Paid and Self-Reported Paid tracking labels, with the notice “Provider payment not verified.” Each utility mark shall belong to the reporting Tenant and the corresponding bill or allocated share, without authorizing a system-issued receipt or changing other tenants' marks. Tenants shall access their bills, history, rent statuses, personal utility tracking, and confirmed rent receipts through the authenticated portal. Eligible reminders shall be scheduled at 9 AM Philippine time, three calendar days before the actual due date. Routine valid billing records shall proceed automatically, while exceptions shall require Manager intervention. SMS, GSM, and Tenant email notifications shall not be used; the mailbox shall serve utility intake only.

B. User Requirements

The Admin shall perform only the implemented system-maintenance functions. The Manager shall log in with valid credentials and manage properties, units, tenants, utility providers, account mappings, rent, billing arrangements, and applicable rules. The

Manager may upload supported bill images when needed. The Manager shall review OCR and billing results, resolve exceptions, and submit corrections for revalidation and recomputation. The Manager shall confirm or reject Tenant-reported rent payments, record and confirm actual rent payments without prior reports after external checking, approve available advance-credit application, and view confirmed rent receipts. The Manager shall not verify utility-provider payments or change a Tenant's personal utility tracking mark. The Tenant shall log in with valid credentials and view assigned property or unit information, rent and utility bills, categories, periods, amounts due, and historical records. Reporting rent payment shall change it to Tenant Reported Paid; Manager confirmation shall mark it Paid, while rejection with a recorded reason shall return it to Unpaid. For utilities, the Tenant may mark their own bill or allocated share as Self-Reported Paid and return it to Not Marked Paid when necessary. These changes shall retain dated history and display “Provider payment not verified.” Tenants shall view confirmed rent receipts and receive permitted Web Push alerts linking to the authenticated portal.

C. Security Requirements

The system shall require valid credentials, restrict functions and records according to the Admin, Manager and Tenant roles, and limit property, tenant, utility, billing, payment, exception, correction, receipt, notification and source-document access to authorized users and functions. Authentication, defined permissions and session

management shall protect records from unauthorized access or modification. User-provided and extracted information shall be validated before storage or processing; failed field, identification, mapping, duplicate or other required checks shall route records to the Exception Queue. Correction information shall be retained in the centralized database. These requirements cover the implemented controls and evaluation criteria and do not constitute legal certification or a complete privacy audit.

2.1 Concept of the Study

Architectural Framework

The framework connects property and unit management, Tenant management, utility-provider and account mapping, and rent configuration with email intake, optional Manager bill uploads, OCR, validation, rule-based classification, duplicate checking, billing computation, and the Exception Queue. Billing records, rent payment-status management, advance-credit application, confirmed rent receipts, personal utility tracking, and Web Push notifications use the Centralized Database. The Admin, Manager, Tenant, and Utility Provider interact with these components according to their defined responsibilities.

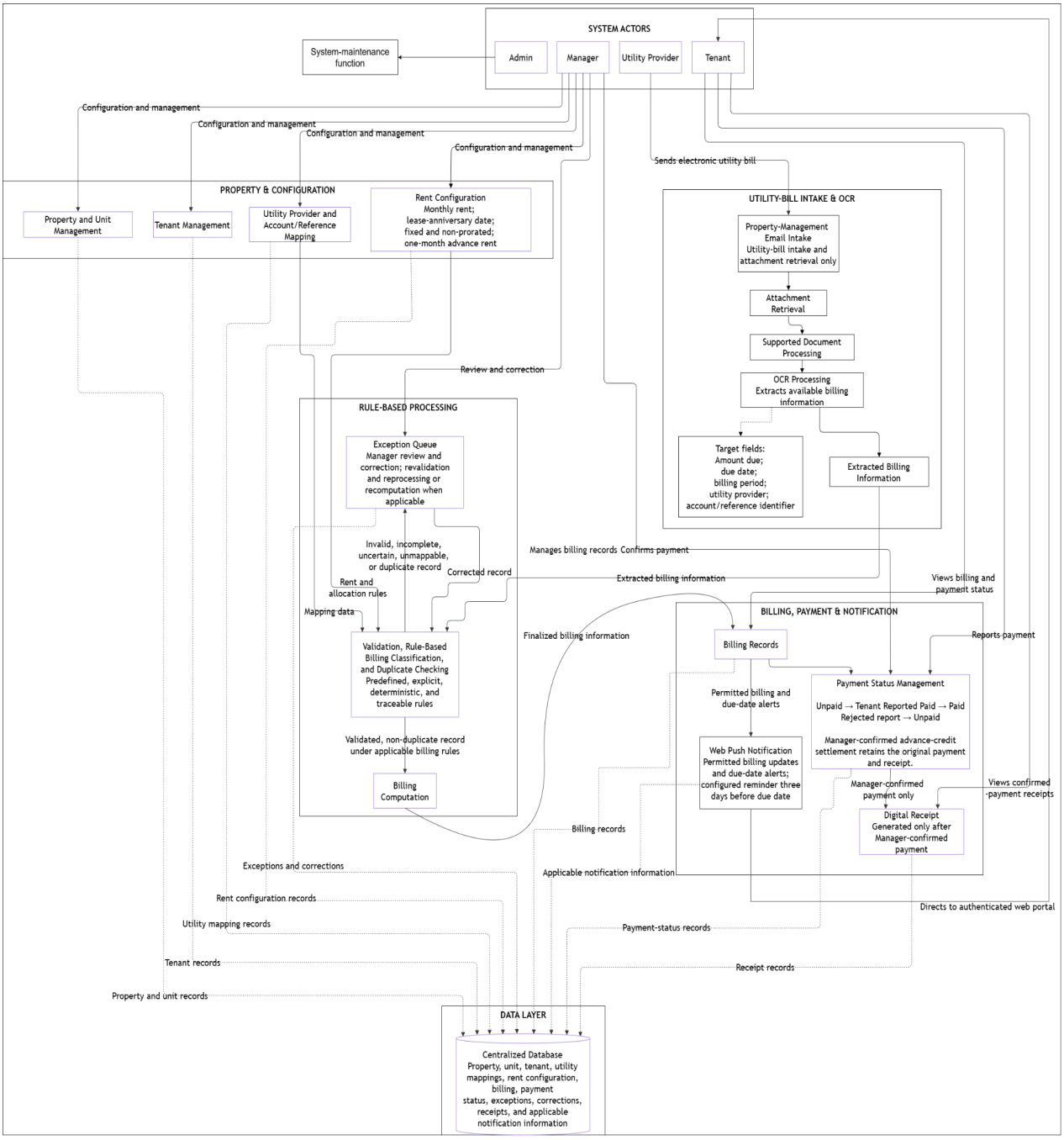


Figure 2.1: Architectural Framework



Property and Unit Management manages property and unit information. Tenant Management manages Tenant information and its association with the applicable property or unit. Utility Provider and Account/Reference Mapping stores the provider, account or billing number, covered property or unit, and Manager-selected Individual or Shared-Occupancy Billing Arrangement. Individual utility bills follow the assigned account without splitting. Shared bills are divided equally among eligible tenants in the configured group. Tenants decide which person pays the whole shared provider bill outside the system; the allocation does not represent collection or verified payment.

Rent will be generated once per monthly lease cycle on the configured anniversary date, which will also be its due date. The charge will be fixed and non-prorated, with its coverage period recorded. If the anniversary day is absent, the system will use the month's last day and retain the original day for later months; a January 31 anniversary therefore uses February 28 or 29 and then March 31. If the server is unavailable, missed eligible cycles will be recovered once when processing resumes. Each recovered bill will retain its original due date, coverage period, and scheduled eligibility date while recording its actual creation date. Generation checks will prevent duplicate cycles, repeated advance-rent lines, and new cycles after the lease end date. Existing unpaid charges will remain accessible, and later membership changes will not silently replace finalized bills. Rental records will use a start date and an optional end date for month-to-month occupancy. A blank end date will mean that the rental continues; it will not require a separate renewal transaction. When the Manager records the Tenant's departure date, rent generation will stop for cycles scheduled after that date.

Previous charges and their associated tracking, payments, and receipts will be retained. The study will not include automatic transfers between units or automatic lease-renewal processing.

The first bill will show current rent and one-month advance rent as separate rent lines. Advance rent is not a security deposit and will become available credit only after Manager confirmation of its payment. Shared tenants will have separate current-rent and advance-rent shares. Available advance credit will be reserved for the final applicable rent cycle and applied only after Manager confirmation. The system will display the final-cycle charge, confirmed available credit, and remaining balance before application. If the approved credit fully covers the charge, the bill will become Paid using the original confirmed payment and receipt reference, without another cash payment or duplicate receipt. Applied credit shall not exceed the available confirmed amount and shall not be reused. Fully consumed credit will no longer be available for another charge, and no replacement advance charge will be generated automatically. Only a remaining outstanding balance may be reported for payment. Credit shall not be applied to a bill already marked Paid; a pending rent report shall require Manager review before application. A mismatch between available credit and the final-cycle charge will require Manager review rather than an invented refund or automatic settlement.

The designated property-management mailbox will retrieve supported utility-bill attachments and will not send Tenant notifications. The Manager may also upload a clear photograph or scan of a printed provider bill. Both intake methods will follow the same

OCR, validation, mapping, and duplicate-checking workflow. An existing OCR engine will recognize text from selected provider layouts in digitally generated PDFs and clear images without researcher training or customization of a machine-learning model. Predefined extraction rules will obtain the amount due, due date, billing period, provider, and account or reference identifier while retaining the source reference. OCR will not establish payment status, billing validity, or Tenant charges; extracted fields will undergo validation and mapping. Actual validated utility due dates will be used for billing and reminders, without predicting dates for documents not yet received.

Absent, unreadable or conflicting values, and unsupported attachments, will require exception review rather than finalized charges. Supported layouts, file types and processing limits will be documented before final testing. Evaluation will compare extracted fields with manually checked samples and report results only within the declared document boundaries.

Validation will check required fields, currency amounts, valid dates and a billing period whose beginning does not follow its end. Account identifiers will remain text to preserve leading zeros. A unique Manager-configured provider/account mapping will identify electricity or water and the covered property or unit; its saved configuration will determine the billing arrangement. Provider aliases will follow an explicit list. Missing, conflicting, invalid or unmappable values will enter the Exception Queue rather than being guessed. Rent will follow saved configuration without OCR classification.

Duplicate checks will compare attachments and provider/account/period combinations with existing records. Repeated or revised bills will require Manager review before another charge or correction is accepted, and reprocessing an accepted source will not create duplicate allocations. Each result will retain its source reference, normalized fields and applied rule identifiers. Corrected records must pass the same checks before computation and finalization.

Table 2.1: Validation and classification rules

Rule ID	Rule Name	Description	Action/Result
V01	Required information	Amount, due date, billing period, provider, or account identifier is missing or conflicting	Exception Queue; no finalization
V02	Valid values	Amount or date cannot be interpreted, amount is negative, or period begins after it ends	Exception Queue; Manager review
V03	Unique mapping	Provider/account identifies exactly one configured target and category	Continue; otherwise Exception Queue
V04	Duplicate check	Identical attachment or same provider/account/period is already recorded	Hold for duplicate or revised-bill review

Rule ID	Rule Name	Description	Action/Result
C01	Utility category	Configured provider/account mapping identifies electricity or water	Assign mapped category
C02	Billing arrangement	Target has a valid individual/shared configuration	Apply saved arrangement; otherwise Exception Queue
A01	Allocation	Eligible group and reconciled shares are available	Proceed to finalization; otherwise Exception Queue
R01	Rent cycle	Lease/cycle already has a rent charge	Do not generate a second charge

Individual charges will follow the configured lease or utility account. Shared rent and utility charges will be split equally among eligible tenants in the Manager-configured property or unit group, without consumption estimates or meter readings. Eligibility will use active assignments on the scheduled rent-cycle date for rent and on the source period's final day for utilities, excluding assignments already ended. Recovered rent cycles will use their original scheduled eligibility dates. The system will retain the eligible list so later occupancy changes do not alter finalized charges. Missing groups, periods, or eligible tenants will enter the Exception Queue. Computation will divide the source charge in whole centavos, assigning equal base shares and distributing leftover

centavos in ascending permanent Tenant-record identifier order. PHP 100.00 split among three tenants will produce PHP 33.34, PHP 33.33, and PHP 33.33. Shares will display two decimal places and must total the source charge before finalization. For shared utilities, tenants will arrange one person to pay the whole provider bill outside the system. Personal tracking will refer only to each Tenant's allocated share and will not establish payment of the full provider account.

The Exception Queue will block finalization until Manager review and successful validation. Corrections will retain the affected field, previous and new values, reason and Manager identity, then undergo revalidation and applicable reprocessing or recomputation. Unresolved records will remain blocked with their failure reason; submitting a correction alone will not establish validity.

Billing Records will retain rent and utility charges while separating rent confirmation from personal utility tracking. Rent will use Unpaid, Tenant Reported Paid, and Paid. Tenants may report full payment of a selected outstanding rent balance; partial payments and overpayments will not be processed automatically. Only one pending rent report shall be retained per bill, and repeated submissions shall not create another report. Tenant reports will remain pending until the Manager checks the payment outside the system. Rejection will record a reason, restore Unpaid, and preserve report history. The Manager may also record and confirm an actual rent payment without a prior Tenant report after external checking. Either confirmation path will retain the confirmed amount,



payment date, Manager identity, and unique receipt. Repeated submission or confirmation of the same transaction will not create another payment or receipt.

Electricity and water will use the personal tracking labels Not Marked Paid and Self-Reported Paid, accompanied by “Provider payment not verified.” Tenants may change only their own bill or allocated-share mark, with dated history retained. In shared arrangements, tenants will decide which person pays the provider bill; a personal mark will not change other tenants' marks or establish settlement of the entire account. Utility tracking will require neither Manager confirmation nor a system-issued receipt. A mark will refer to the recorded bill or allocation version, so later charge corrections will preserve earlier history and require Tenant review of the revised information. The system will not independently verify cash, bank transfers, electronic-wallet transactions, Tenant contributions, or utility-provider payments.

Finalized corrections will preserve prior values and processing history. Unpaid rent charges may be corrected after revalidation and recomputation; pending reports must be reviewed against revised amounts. Paid rent bills will retain their original payment and receipt, with corrections recorded as linked revisions showing charge and balance changes. Utility corrections will retain their original source, allocation, and associated personal tracking history and undergo applicable validation and recomputation. External payments or refunds will remain outside automated settlement. Tenants will access current and historical records through the authenticated portal.

The Centralized Database will relate properties, units, tenants, leases, utility mappings, billing sources, allocations, rent payment reports, Manager-recorded payments and confirmations, confirmed rent receipts, advance-credit applications, utility tracking history, exceptions, corrections, and notifications. Recorded assignments will support multiple tenants per shared unit. Allocations will link to their source and eligible group; rent payments and receipts will link to their bills. Personal utility tracking will link to the assigned Tenant and the relevant bill or allocated share. These relationships will preserve transaction history and support access restrictions. Corrections will remain centralized records rather than a separate architectural component. The database design will define storage relationships while the framework, flowchart, and DFD describe structure and operation.

Web Push will be the only external Tenant-notification channel. On a supported browser, Tenant permission will allow creation and storage of an account/device subscription and the background handling needed to receive an alert. Alerts will open the authenticated billing record without revealing complete details beforehand or confirming payment. Finalization and relevant corrections will trigger billing updates. Reminders will be scheduled at 9 AM Philippine time, three calendar days before the actual due date, for eligible Unpaid rent and utility records Not Marked Paid. Rent reports awaiting Manager confirmation and utility records marked Self-Reported Paid will be excluded. A utility reminder will prompt the Tenant to check whether the bill or share has been paid; suppressing it after a personal mark will not establish provider settlement. Corrected due dates will replace pending schedules, and repeated events will not duplicate reminders.

Bills unavailable before the scheduled reminder time, including same-day rent and late recovered bills, will receive billing updates rather than backdated reminders. If the server misses the reminder time, it will not send a retrospective reminder for that missed event.

Notification records will distinguish scheduling, attempts, and available push-service acceptance or failure results. Acceptance will not prove receipt or reading. Invalid subscriptions, denied permission, unsupported devices, and delivery failures will not prevent authenticated access to bills, rent statuses, personal utility tracking, or confirmed rent receipts. SMS, GSM, and Tenant email will remain excluded.

The Admin performs system-level maintenance. The Manager manages property information, Tenant records, utility mappings, billing configurations, bill uploads, exceptions, corrections, rent payments and confirmation, approved advance-credit application, and confirmed rent receipts. The Tenant views their billing information and history, reports rent payments, maintains their own utility tracking marks, views confirmed rent receipts, and receives permitted Web Push Notifications. The Utility Provider supplies electronic bills received through the designated property-management email account or printed bills submitted as clear images by the Manager. Provider settlement and the arrangement of shared-bill contributions remain outside the system's verification functions.

The components form one web-based workflow. Saved property, Tenant, mapping, and rent information supports utility intake, OCR, validation, allocation, and exception handling. Rent payment confirmation, final-cycle advance credit, confirmed

rent receipts, and personal utility tracking remain separate functions. The Centralized Database retains their records, while Web Push provides permitted billing and due-date alerts. Each actor uses only the functions assigned to their role.

2.2 Conceptual Operation

The workflow uses property, Tenant, utility-mapping, and rent configuration to process rent, electricity, and water charges. Monthly rent follows the fixed anniversary rule, with one-month advance rent shown separately on the first bill. Missed eligible cycles are recovered once with original due dates and actual creation dates retained. Utility processing receives property-email attachments or Manager-uploaded clear images of printed bills, extracts their fields through OCR, and validates required information, mappings, periods, and duplicates. Valid records proceed to classification, computation, and finalization; exceptions require Manager correction and renewed validation.

Shared charges use equal splitting among configured eligible tenants, using the scheduled rent-cycle date or the utility period's final day. Whole-centavo computation and consistent remainder distribution preserve the source total. Rent begins as Unpaid; Tenant reports become Tenant Reported Paid until Manager confirmation, while rejected reports return to Unpaid with recorded reasons. The Manager may also record and confirm externally checked rent payments without a Tenant report. Confirmed advance credit may cover the final applicable rent bill with Manager-confirmed application, reusing

the original payment and receipt rather than recording another cash transaction. Consumed credit cannot be reused, and no replacement advance is charged automatically. Utilities use separate Tenant-controlled Not Marked Paid and Self-Reported Paid marks with “Provider payment not verified.” Tenants arrange the payer of a shared provider bill outside the system; individual marks do not confirm payment of the full account or authorize receipts. Permitted billing updates and eligible reminders at 9 AM Philippine time, three calendar days before the actual due date, link to the authenticated portal without proving payment, delivery, or reading.

Process Flowchart

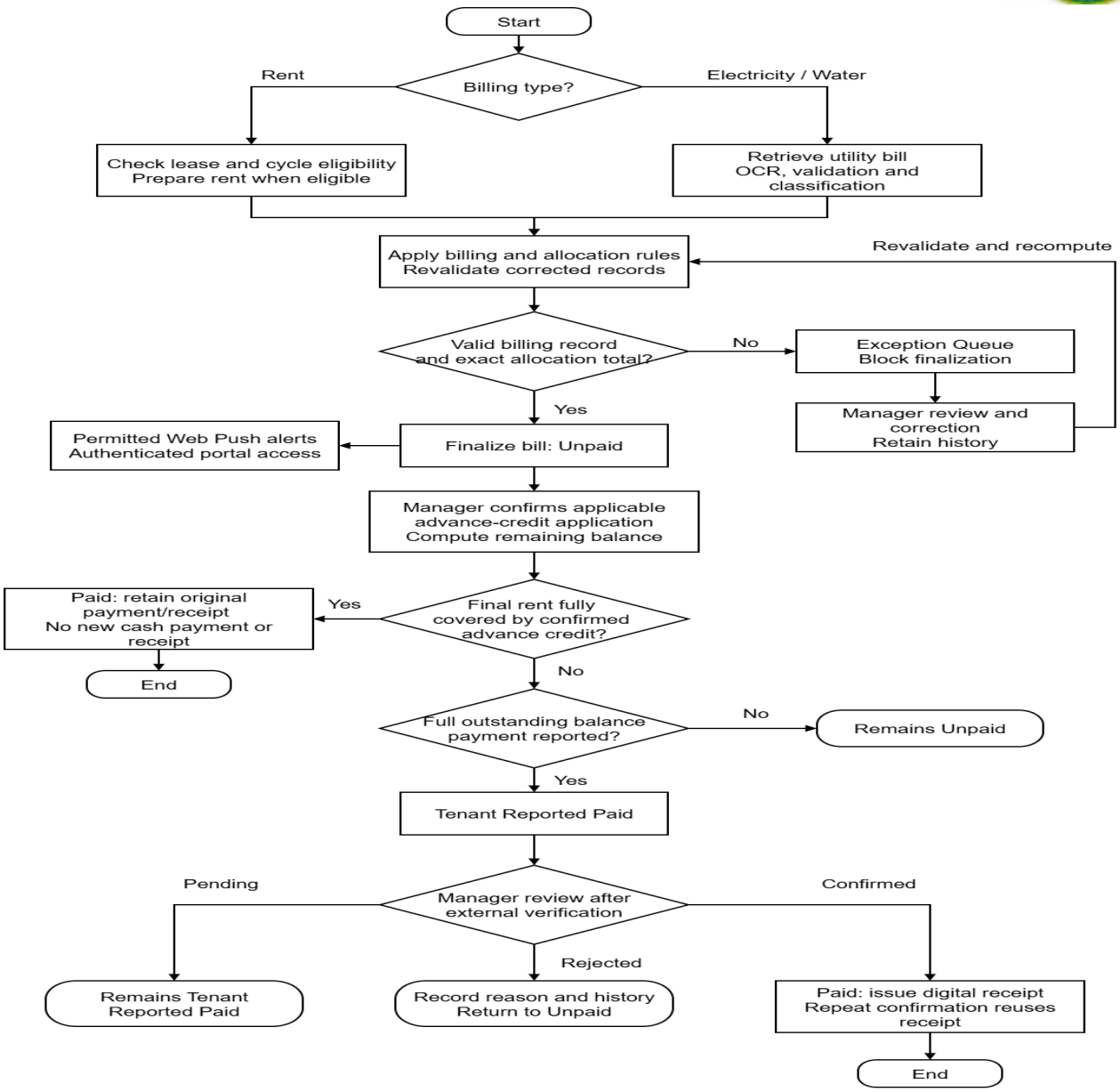


Figure 2.2. Overall Process Flowchart

The flowchart shows rent and utility intake, validation, allocation, and finalization, with failed records returned for Manager correction and revalidation. Rent continues through Tenant reporting or direct Manager recording, confirmation, and confirmed rent

receipts; final-cycle advance credit reuses the original transaction references. Utilities continue through separate personal tracking without Manager payment confirmation or system-issued receipts. Web Push supports both billing paths through permitted alerts.

Full-Blown Data Flow Diagram

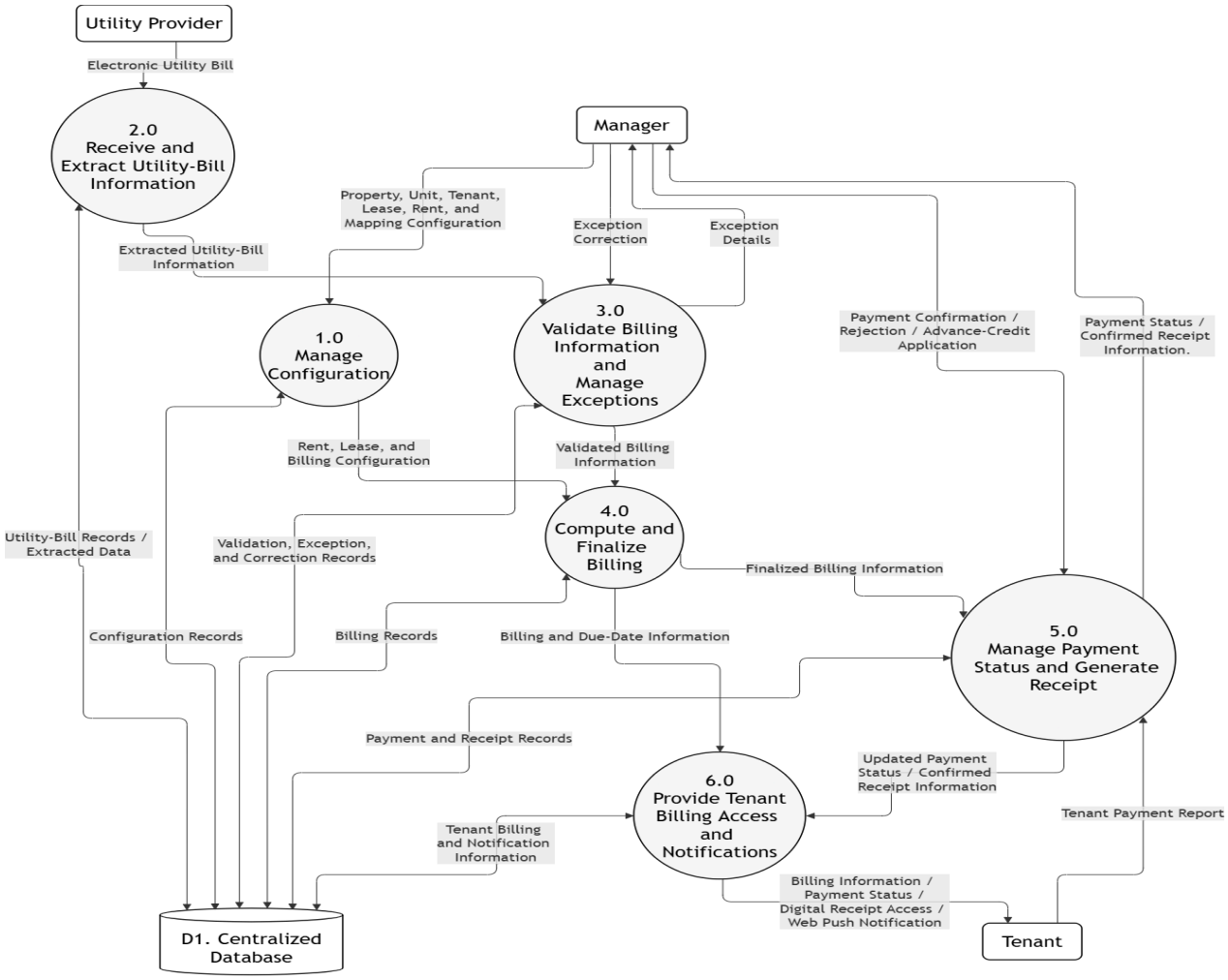


Figure 2.3. Data Flow Diagram

The DFD illustrates how billing-related information moves among the Manager, Tenant, Utility Provider, system processes, and Centralized Database. It shows the flow of configuration data, utility-bill information, billing records, payment information, receipt information, and tenant notification information through the system processes. External entities interact with the system through the appropriate processes, while the Centralized Database stores and provides the records required by the system.

2.3 Mockup Design

The mockups present the proposed Admin, Manager, and Tenant interfaces. Admin screens cover the included maintenance functions. Manager screens show property records, billing configuration, utility intake, exceptions, rent payment recording and confirmation, and final-cycle credit application. Tenant screens show assigned bills and history, rent reporting, confirmed rent receipts, personal utility tracking, and permitted notifications. These designs guide development and do not represent completed or evaluated functions. Their actions and status labels will follow the approved workflow.

2.4 Methodology of the Study

The study will use the Iterative Software Development Methodology to develop the proposed Intelligent Property Management Workflow with OCR-Enabled Automated

Billing Classification and Tenant Notification. This methodology will organize development into successive iterations in which related system functions will be planned, designed, developed, reviewed, refined, and integrated.

The methodology is appropriate for the proposed system because its functions are interconnected but can be developed in manageable groups. These functions include system foundation, property and billing configuration, utility-bill intake, OCR extraction, rule-based processing, exception handling, Billing Computation, payment and receipt management, tenant access, Centralized Database functions, and Web Push Notifications. The methodology will also support the intended automate by default, intervene by exception approach.

2.4.1 Principles of the Iterative Methodology

The Iterative Software Development Methodology will be guided by four development principles and the system's exception-based workflow. First, incremental development will allow the researchers to develop related system functions in smaller iterations instead of implementing the entire system at once. Second, continuous testing and review will allow each iteration to be checked against the defined requirements, user roles, workflow rules, and data requirements.

Third, refinement between iterations will allow issues identified during development to be corrected before related functions are integrated into succeeding iterations. Fourth, cumulative integration will allow functions developed in earlier iterations to work with functions developed later until they form a complete system.

Throughout the iterations, development and testing will preserve the system's exception-based workflow, allowing valid records to proceed while records requiring review enter the Exception Queue for Manager intervention.

2.4.2 Development Iterations of the Methodology

The proposed system will be developed through seven planned iterations based on the functional requirements of the study. Each iteration will focus on a related group of functions that will later be integrated into the complete system.

Table 2.2: Development Iterations of the Proposed System

Iteration	**Development Focus**	**Main Functions**
Iteration 1	Requirements and System Foundation	User roles, authentication, property, unit, tenant, and Centralized Database structure
Iteration 2	Property and Billing Configuration	Utility providers, account or reference mapping, rent configuration, billing arrangements, and billing rules
Iteration 3	Property-Management Email Intake, Bill Upload, and OCR	Designated email intake, attachment retrieval, optional Manager upload of clear printed-bill images, supported document processing, and OCR extraction
Iteration 4	Rule-Based Validation and Exception Handling	Required-field checking, mapping, billing-period checking, duplicate checking, billing classification, Exception Queue, Manager correction, and revalidation

Iteration	Development Focus	Main Functions
Iteration 5	Billing Computation and Finalization	Rent, electricity, and water billing; Individual and Shared-Occupancy Billing Arrangements; allocation; computation; and Billing Finalization
Iteration 6	Rent Payment, Utility Tracking, Digital Receipt, and Web Push Notification	Rent statuses, Tenant rent reporting, Manager-recorded payments and confirmation, final-cycle advance-credit application, confirmed rent receipts, personal utility tracking, Tenant portal access, and Web Push alerts
Iteration 7	System Integration and Testing	Integration of system functions, implemented security controls, system-wide testing, and final evaluation

2.4.3 Application of the Methodology to the Study

The planned iterations will be applied progressively, beginning with the system foundation and configuration functions and continuing with property-email intake, optional Manager bill uploads, OCR extraction, rule-based processing, exception handling, Billing Computation, rent payment and receipt management, approved advance-credit application, personal utility tracking, Tenant access, and Web Push Notifications. Each iteration will build upon related functions developed in earlier iterations.

During development, each iteration will undergo planning, design, development, testing and review, followed by refinement and integration. Findings from testing and

review will be used to correct related functions before succeeding components are integrated. Records requiring Manager intervention will be handled through the Exception Queue according to the defined processing rules.

The final iteration will integrate the developed functions and conduct system-wide testing and evaluation. The completed system will be evaluated using the selected ISO/IEC 25010:2023 quality characteristics: Functional Suitability, Performance Efficiency, Interaction Capability, Reliability, and Security. No completed iteration, testing result, or evaluation result is claimed in this chapter.

This approach will organize related functions into manageable development cycles. Repeated testing and review will help identify defects and maintain consistency among requirements, roles, workflows, and data structures. Gradual integration will connect these functions while preserving the intended automate-by-default, intervene-by-exception approach.

2.4.4 System Testing and Evaluation Procedures

Testing will occur during each iteration and after integration, using business, user and security requirements. Cases will cover normal transactions, incomplete fields, unmapped accounts, duplicates and corrected exceptions, recording inputs, expected and observed results, and pass or failure. Defects will be corrected and affected tests repeated before final evaluation. OCR results for the five target fields will be compared



with manually checked documents. Field accuracy will equal correctly extracted values divided by expected values, multiplied by one hundred; missing and incorrect values will be recorded. Declared normalization rules must preserve meaning. Document sets and rules will be fixed before final testing, with refinement samples identified separately from final evaluation samples.

Validation, mapping, classification, and exception routing will be checked against predefined outcomes for both email attachments and Manager-uploaded printed-bill images. Rent and utility computations will be checked independently, including individual shares, exact source reconciliation, anniversary dates, short-month handling, and missed-cycle recovery without duplicate charges. Rent payment tests will verify Tenant reporting, direct Manager recording after external checking, Manager-only confirmation, approved advance-credit use, remaining-balance reporting, and original-receipt reuse without repeated credit or automatic replacement charges. Utility tracking tests will verify that only the assigned Tenant may change their own mark, dated history is retained, and no mark establishes provider settlement or generates a system-issued utility receipt. Reminder tests will check the 9 AM Philippine-time cutoff, three-calendar-day timing, late bills, pending rent reports, and self-reported utility marks. Repeated processing will be checked for duplicate charges, payments, credit applications, receipts, or reminders.

Functional Suitability will use requirement tests; Performance Efficiency will use processing and response measurements under a stated environment; Interaction Capability will use representative tasks and an adviser-reviewed questionnaire;

Reliability will use repeated-processing and recovery tests; and Security will use authentication, role and unauthorized-record-access tests. User ratings will supplement rather than replace technical evidence.

Managers will review bills, correct exceptions, record or confirm rent payments, and apply eligible final-cycle credit. Tenants will view bills and history, report outstanding rent, use personal utility tracking, and access confirmed rent receipts. Task completion and difficulties will be recorded. Questionnaire responses will use the approved scale and scoring procedure, while technical results will use counts, percentages, time summaries, and failed-case descriptions. Evaluator selection and counts, document counts, and questionnaire scoring will be settled with the adviser before final data collection. No completed testing or evaluation result is claimed at the proposal stage.